BOSTON UNIVERSITY

Milestone Three: Exploratory Data Analysis (EDA)

Identifying 30-Day Readmission Risk in Diabetic Hospital Encounters

This document summarizes the exploratory data analysis completed across Weeks 1 through 7 for the Chronic Disease, Hospital Readmission & Population Health Outcomes project. It covers what was learned through preprocessing, univariate, and bivariate analysis of the Diabetes 130-US Hospitals dataset, the conclusions drawn about data quality and usability, and the supervised and unsupervised methods best suited for predicting 30-day readmission in the next phase of the project.

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DX699 O2 AI for Leaders (Fall 26)

1. Description of Project

This project focuses on the healthcare industry, specifically the relationship between diabetes, hospital utilization, and readmission. The chosen project, **Chronic Disease, Hospital Readmission & Population Health Outcomes Analysis**, brings together 4 public datasets that look at chronic disease from different levels: the individual patient, the hospital encounter, the general population, and the country as a whole (Table 1). The primary dataset for this analysis is **Diabetes 130-US Hospitals for Years 1999 to 2008**, which contains 101,766 inpatient encounters for diabetic patients across 130 U.S. hospitals (Clore et al., 2014). Each row represents 1 hospital stay and includes demographics, admission and discharge details, diagnosis codes, lab results, 23 diabetes medication fields, prior visit counts, and a readmission outcome.

The central question guiding this analysis is which patient characteristics and utilization patterns are associated with readmission within 30 days, and is the data strong enough to support a predictive model? This question carries over from my previous group's Milestone Two work last semester, where readmissions were framed as a measurable indicator of avoidable cost and inefficiency in care delivery. For Milestone Three, the focus shifts from describing the datasets to actually exploring one of them in depth.

30 day readmissions matter because they are one of the most widely used benchmarks for hospital quality and cost. Under Medicare's Hospital Readmissions Reduction Program, hospitals with higher than expected readmission rates can see reduced payments (Centers for Medicare & Medicaid Services, 2024). That puts the primary audience for this work in 2 places: insurers like Medicare who want to control unnecessary spending, and the hospital care teams responsible for discharge planning. If high-risk patients can be identified before they even leave the hospital, they can be prioritized for follow-up calls, medication reviews, and earlier outpatient visits. The

potential impact is lower avoidable cost for insurers and patients, fewer penalties for hospitals, and better outcomes for a population that is already heavily hospitalized.

Table 1. Datasets in the chosen project and utilization

Dataset	Source	Unit of analysis	Used in this paper
Diabetes 130-US Hospitals	UCI	1 hospital encounter	Yes, primary dataset
SUPPORT2	UCI	1 critically ill patient	Set aside
Diabetes Health Indicators	Kaggle	1 survey respondent	Set aside
Life Expectancy	Kaggle	1 country-year	Set aside

2. What I Learned from EDA

The exploratory work for this project was completed in 3 passes across the weekly homework. Week 2 focused on cleaning and preprocessing, Week 4 looked at each column on its own, and Week 6 looked at how columns relate to each other and to the outcome. The odd weeks (1, 3, and 5) practiced the same skills on smaller practice datasets first, which made it easier to apply them to the larger hospital data. Throughout the analysis, the target variable is **readmitted**, which has 3 values: NO, >30 days, and <30 days.

2.1 Preprocessing

The first thing preprocessing revealed is that this dataset hides its missing values. Instead of leaving fields blank, missing values were stored as a question mark, so a standard null check initially showed almost nothing missing. Once the question marks were converted to true missing values, the picture changed drastically. 3 columns were mostly empty: **weight** (96.9% missing), **medical_specialty** (49.1%), and **payer_code** (39.6%). All 3 were dropped, since filling that many values would essentially mean making up data. **Race** was only missing for 2.2% of encounters, so those rows were kept and labeled "Unknown."

2 lab columns looked just as empty but needed a different approach. **A1Cresult** was blank for 83.3% of encounters and **max_glu_serum** for 94.7%. In this case, a blank doesn't

mean the value was lost. It means the test was never ordered. Similar to how in my line of work a blank churn date can mean a customer is still active. The absence of a value here carries meaning on its own, so both columns were filled with "Not tested" and kept as a real category. I also converted admission_type_id, discharge_disposition_id, and admission_source_id from numbers to text, because they are category codes and averaging them wouldn't mean anything.

Next, I removed rows that couldn't be used. 1,522 encounters were missing a diagnosis, 3 had an invalid gender, and 1,650 ended with the patient passing away during the stay. That last group is important, since a patient who died can't be readmitted, and keeping them would quietly pull the readmission rate down. After these steps, 98,591 encounters remained, which is 96.9% of the original data (Figure 1).

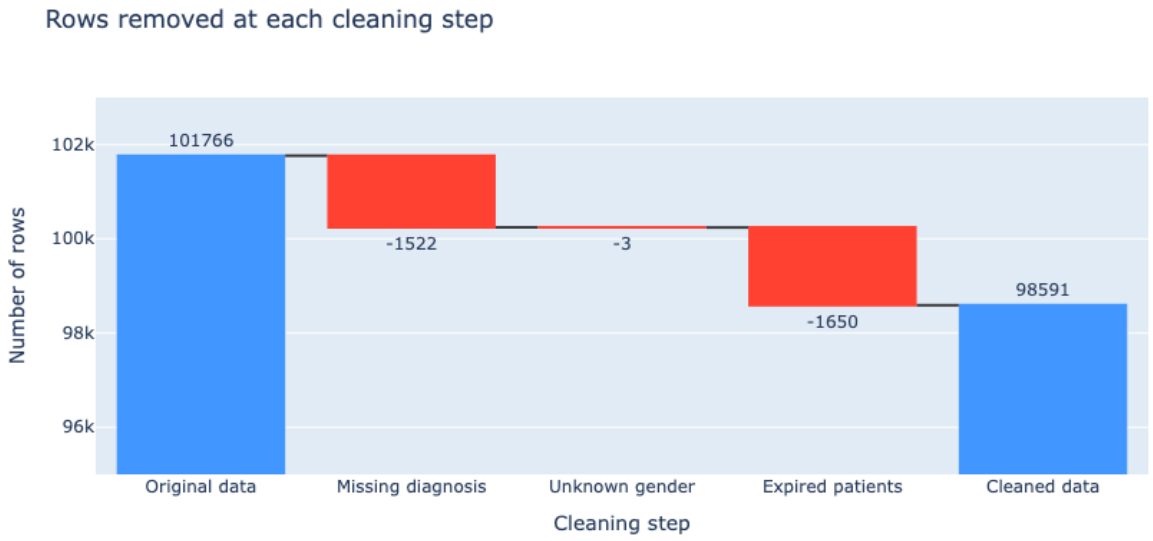


Figure 1. *Rows removed at each cleaning step, 101,766 to 98,591 (Week 6 notebook)*

There were no duplicate rows, and every encounter_id was unique. However, the data only contains 69,333 unique patients, meaning 29,258 encounters are repeat visits by the same people. That makes sense for a readmission dataset, but it also means the rows aren't fully independent of each other, which becomes important later when splitting the data for modeling.

The last major finding from preprocessing was class imbalance (Figure 2). Only 11.4% of encounters were readmitted within 30 days, while 35.7% were readmitted after 30 days and 52.9% were never readmitted. The group that matters most here is also the smallest. This has direct impact on modeling: a model that simply predicted "not readmitted within 30 days" for every patient would be 88.6% accurate while catching 0 high-risk patients. Accuracy alone won't be a useful way to measure success here.

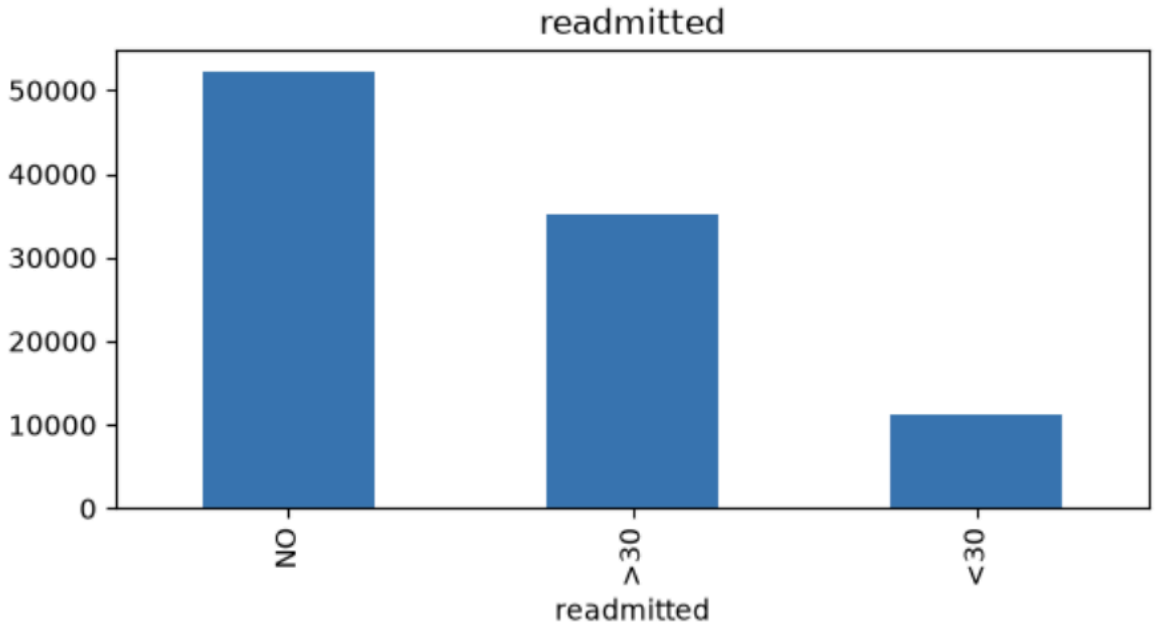


Figure 2. *Counts of the readmitted target after cleaning (Week 4 notebook)*

2.2 Univariate Analysis

Looking at each numeric column on its own showed that very few of them follow a normal distribution (Figure 3). **Days in hospital** (mean 4.4, max 14), **number of medications** (mean 16.1, max 81), and **number of procedures** are all skewed to the right, meaning most stays are short and a smaller number run long. That lines up with what I would expect from hospital

data.

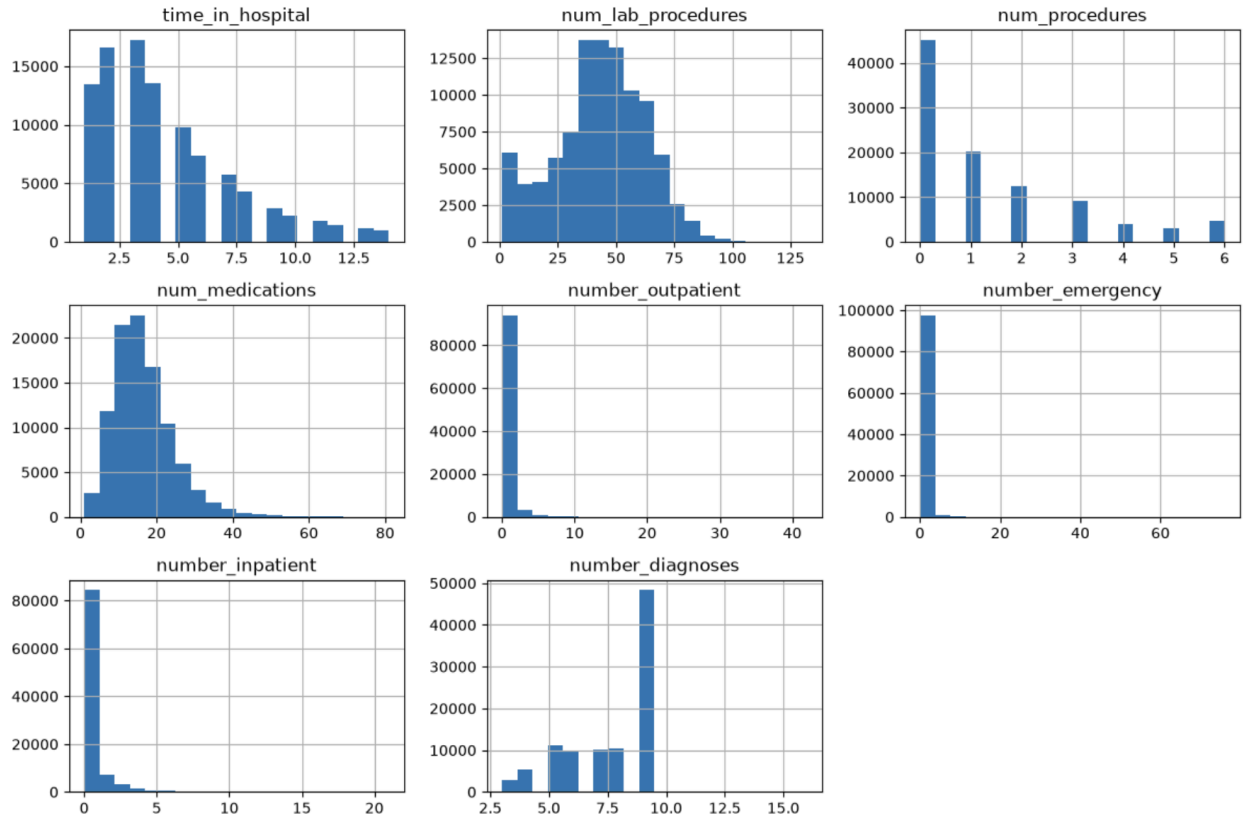


Figure 3. *Histograms of the 8 numeric columns (Week 4 notebook)*

The 3 prior visit columns were the most extreme. In the year before the encounter, 83.4% of patients had no outpatient visits, 88.7% had no emergency visits, and 66.4% had no inpatient visits. At the same time, the maximums reach 42 outpatient, 76 emergency, and 21 inpatient visits. I looked into these and decided to keep them. They look like real frequent patients of the hospital rather than data entry errors, and those patients are exactly the type I am trying to identify.

2 columns showed patterns that seem to come from how the data was recorded rather than actual patient behavior. **Number of diagnoses** piles up at 9, with only 111 encounters above it, which suggests most hospital systems in the dataset only stored up to 9 diagnoses during this period. **Number of lab procedures** looked like a clean bell curve with 10 bins, but switching to

50 bins revealed a spike of 3,126 encounters with exactly 1 lab procedure, which is also the mode (Figure 4). This was a good reminder that bin size can hide important details. The spike could be a documentation shortcut or a different type of stay, but either way it would have been missed using the default settings.

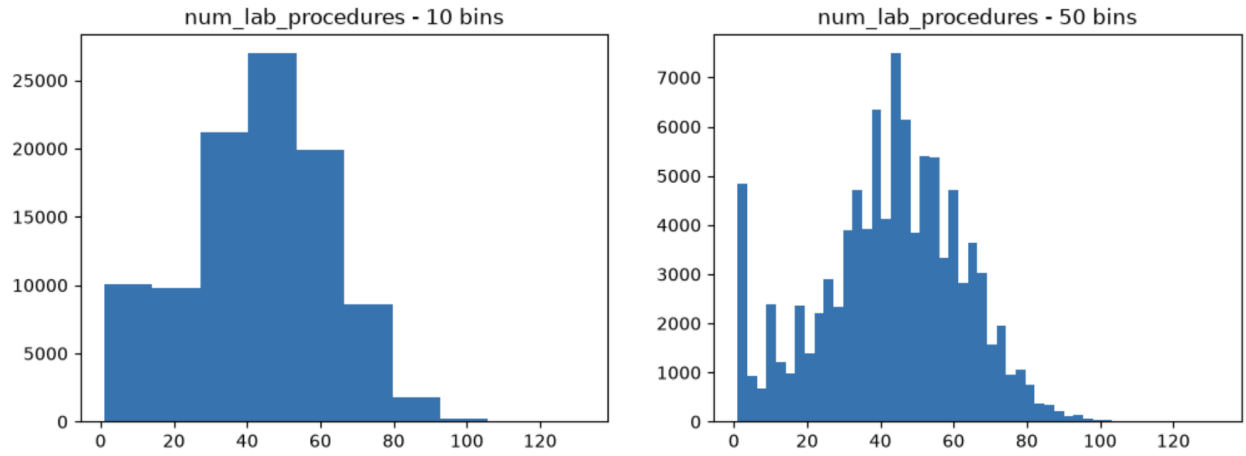


Figure 4. *num_lab_procedures at 10 bins and 50 bins (Week 4 notebook)*

The categorical columns describe an older patient population. 64.9% of encounters were for patients between 60 and 89, with the 70 to 79 age group being the largest. The data is 74.9% Caucasian and 18.8% African American, so results may not generalize well to other groups. The most surprising single number in the dataset was that 83.3% of encounters had no A1C test recorded. A1C is the main measure of long-term blood sugar control for diabetic patients, so I expected it to be tested far more often.

2.3 Bivariate Analysis

The strongest relationship in the data is between prior hospital use and readmission (Figure 5). Patients with no inpatient visits in the previous year were readmitted within 30 days 8.6% of the time. That rate increases at every step, reaching 31.2% for patients with 4 or more prior inpatient visits, which is about 3.6 times higher. The group averages show the same pattern. Patients readmitted within 30 days averaged 1.22 prior inpatient visits, compared to 0.84 for

those readmitted after 30 days and 0.37 for those never readmitted.

```
Readmission percentages by prior inpatient visits (4 = 4 or more):
readmitted          <30   >30    NO
number_inpatient
0                    8.6  30.7  60.7
1                   13.2  42.6  44.2
2                   17.8  48.8  33.4
3                   20.9  50.7  28.4
4                   31.2  49.4  19.4

Ratio of <30 rate for 4+ prior visits vs. 0 prior visits: 3.6
```

Figure 5*. Readmission percentages by prior inpatient visits, 0 to 4 or more (Week 6 notebook)*

The correlation table puts that finding in perspective (Figure 6). The strongest correlations in the dataset are between columns that describe the size of a hospital stay: days in hospital and number of medications (0.46), number of procedures and number of medications (0.38), and days in hospital and number of lab procedures (0.32). None of these are high enough to make 1 column redundant, so all of them can stay in the analysis. When compared to 30-day readmission, prior inpatient visits has the highest correlation at only 0.17. Prior emergency visits follows at 0.06, and everything else is 0.05 or lower. The pair plot from Week 6 tells the same story, with the 3 readmission groups overlapping in nearly every scatter plot.

	time_in_hospital	num_lab_procedures	num_procedures	num_medications	number_outpatient	number_emergency	number_inpatient	number_diagnoses	readmitted_30
time_in_hospital	1.00	0.32	0.19	0.46	-0.01	-0.01	0.07	0.21	0.04
num_lab_procedures	0.32	1.00	0.05	0.27	-0.01	-0.00	0.04	0.15	0.02
num_procedures	0.19	0.05	1.00	0.38	-0.03	-0.04	-0.07	0.06	-0.01
num_medications	0.46	0.27	0.38	1.00	0.04	0.01	0.06	0.24	0.04
number_outpatient	-0.01	-0.01	-0.03	0.04	1.00	0.09	0.11	0.09	0.02
number_emergency	-0.01	-0.00	-0.04	0.01	0.09	1.00	0.27	0.06	0.06
number_inpatient	0.07	0.04	-0.07	0.06	0.11	0.27	1.00	0.10	0.17
number_diagnoses	0.21	0.15	0.06	0.24	0.09	0.06	0.10	1.00	0.05
readmitted_30	0.04	0.02	-0.01	0.04	0.02	0.06	0.17	0.05	1.00

Figure 6. *Correlation table of the numeric columns and readmitted_30 (Week 6 notebook)*

Length of stay has a more modest relationship with readmission. The 30-day readmission rate rises from 8.5% for 1-day stays to 14.7% for 10-day stays, then levels off around 13% to

14%. Longer stays do carry slightly more risk, but not nearly as much as a history of prior admissions.

Some of these relationships are likely influenced by confounding variables. Illness severity isn't measured directly in the dataset, but it could explain longer stays, more medications, and more readmissions all at once. Age is another possible factor, since patients 70 and older make up 48.2% of 30-day readmissions compared to 43.7% of encounters that were never readmitted. Repeat patients also play a role, since the same frequent users show up across many rows.

3. Conclusions

Overall, the data is usable and the main patterns make sense, but the EDA uncovered several issues that need to be accounted for before moving into modeling.

3.1 Expected and Unexpected Aspects of the Data

Most of the largr patterns lined up with what I expected going into the analysis. Several of the smaller details did not, and each one has a practical effect on how the data should be analyzed (Table 2).

Finding	Expected?	Impact on analysis
Mostly older patients with short stays	Expected	Fits a diabetic inpatient population; results apply mainly to older adults
Prior inpatient visits predict readmission (8.6% to 31.2%)	Expected	Strongest single feature; should anchor any model
Only 11.4% are 30-day readmissions	Expected	Requires class weighting and recall-based evaluation
83.3% of encounters have no A1C test	Unexpected	Keep "Not tested" as a category; testing itself may carry signal
Spike of 3,126 encounters at exactly 1 lab procedure	Unexpected	Possible recording artifact; check before modeling
Diagnoses capped at 9	Unexpected	Column reflects recording limits, not true illness burden

Finding	Expected?	Impact on analysis
All numeric correlations with readmission are 0.17 or lower	Unexpected	Numeric columns alone will not be enough to predict readmission
No admission dates for a 10-year period	Unexpected	Changes over 1999 to 2008 cannot be checked

***Table 2**. Expected and unexpected findings and their impact on analysis*

The A1C finding stood out the most because it connects directly to published research on this exact dataset. Strack et al. (2014), the researchers who created the dataset, found that measuring A1C was associated with lower readmission rates. My own results point in the same direction. The 30-day readmission rate is 11.7% when A1C was not tested, compared to about 10% when it was. That supports treating "Not tested" as useful information rather than noise.

3.2 Disqualified Data

Within the project, I kept the Diabetes 130-US Hospitals dataset and set the other 3 aside. None of them were disqualified because of poor data quality. They were set aside because they can't answer a readmission question at the encounter level or be linked to the primary dataset.

SUPPORT2 focuses on critically ill patients and in-hospital survival, which is a different population and a different outcome. **Diabetes Health Indicators** is a population survey with 1 row per respondent and no hospital or readmission fields, so there is no common key to join it to hospital encounters. **Life Expectancy (WHO)** is measured at the country and year level and can't describe individual U.S. patients. The impact is that this analysis can't add lifestyle, behavioral, or population-level context, which the project brief envisions as part of the broader picture. Those datasets may still be useful later as supporting context rather than as modeling inputs.

Within the primary dataset, nothing was disqualified as a whole. 3 columns (weight, payer_code, and medical_specialty) and 3,175 rows were removed, which is 3.1% of the data. Losing **payer_code** is the most costly tradeoff for this audience, since it prevents a direct comparison between Medicare and privately insured patients. Losing **weight** removes a well-known diabetes risk factor, and the lack of admission dates means the results describe the full 1999 to 2008 period rather than how readmission patterns may have changed over time.

3.3 Suggested Supervised and Unsupervised Analyses

The most natural supervised approach is binary classification: will this encounter be followed by a readmission within 30 days? I would start with logistic regression as a baseline model, since its results are easy to explain to an insurer or care team. However, the EDA suggests it will have limited performance on its own. The numeric columns are skewed, their correlations with readmission are weak, and much of the useful information sits in categorical fields like diagnosis codes and the 23 medication columns.

For that reason, I would choose gradient-boosted trees, such as **LightGBM** or **XGBoost**, as the primary model. Tree-based models handle skewed data, nonlinear relationships, and mixed feature types without heavy transformation, and they can produce feature importance scores that show which factors are driving readmission risk. Random forest would be a reasonable alternative to compare against. 3 findings from the EDA would directly shape how the model is trained:

- **Class imbalance:** since only 11.4% of encounters are positive cases, I would use class weights and evaluate the model on recall, precision-recall AUC, and F1 score instead of accuracy.

- **Repeat patients:** since 29,258 encounters are repeat visits, the train and test split should be grouped by patient_nbr so the same patient never appears in both.
- **Diagnosis codes:** since diag_1 through diag_3 contain hundreds of ICD-9 codes, they should be grouped into broader disease categories before modeling.

For unsupervised analysis, I would use k-prototypes clustering, which works with both numeric and categorical data, to group patients based on hospital use and medication patterns. This could surface segments such as frequent hospital users or heavily medicated short-stay patients. Principal component analysis (PCA) could help visualize those clusters, and the cluster labels could be added back as a feature in the supervised model. Overall, the goal is not just to predict who will be readmitted, but to give care teams a clearer picture of which patients need attention before they leave the hospital.

4. Citations / Bibliography

Centers for Medicare & Medicaid Services. (2024). *Hospital Readmissions Reduction Program (HRRP)*. https://www.cms.gov/medicare/payment/prospective-payment-systems/acute-inpatient-pps/hospital-readmissions-reduction-program-hrrp

Clore, J., Cios, K., DeShazo, J., & Strack, B. (2014). *Diabetes 130-US hospitals for years 1999-2008* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5230J

Knaflic, C. N. (2015). *Storytelling with data: A data visualization guide for business professionals*. Wiley.

Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., & Clore, J. N. (2014). Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records. *BioMed Research International, 2014*, Article 781670. https://doi.org/10.1155/2014/781670

5. AI Appendix

I used Claude Opus 5.5 only to proofread my final version of this paper. I asked it to check for spelling, grammar, formatting and clarity issues after the writing was complete. All of the analysis, numbers, writing, and conclusions in this paper are my own, and the AI tool was not used to generate the content.

6. Weekly Graphs and Homework Assignments

All 7 weekly notebooks are attached as separate .ipynb files and are also available in my public repository: https://github.com/justin-bosco/OMDS-ModB2-Homework-Bosco-Justin. The repository includes diabetic_data.csv and the image files the notebooks load.

Notebook	Topic	Data
OMDS-ModB2-Week1-Bosco-Justin.ipynb	Preprocessing	Bank customers
OMDS-ModB2-Week2-Bosco-Justin.ipynb	Preprocessing, part 2	Diabetes 130-US Hospitals
OMDS-ModB2-Week3-Bosco-Justin.ipynb	Univariate analysis	Amusement park rides
OMDS-ModB2-Week4-Bosco-Justin.ipynb	Univariate, part 2	Diabetes 130-US Hospitals
OMDS-ModB2-Week5-Bosco-Justin.ipynb	Bivariate analysis	Pair plot exercises (practice)
OMDS-ModB2-Week6-Bosco-Justin.ipynb	Bivariate, part 2	Diabetes 130-US Hospitals
OMDS-ModB2-Week7-Bosco-Justin.ipynb	Milestone week	Graph question and SWD chart